Summer flounder (*Paralichthys dentatus*)

Summary Scores

- **Sensitivity:** Low
- **Exposure:** Moderate
- **Vulnerability:** Low

Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	1.72	2.8	
Complexity in Reproductive Strategy	2	2.6	
Dispersal of Early Life Stages	1.72	2.8	
Egg and Larval Survival and Settlement Requirements	2.4	2.9	
Historical Variability in Temperature	1.56	2.9	
Juvenile Habitat Requirements	2.68	2.7	
Other Stressors	2.28	2.54	
Population Growth Rate	2.16	2.9	
Prey Specificity	1.44	2.9	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.32	2.9	
Spawning Cycle	1.72	2.56	
Stock Size/Status	2.28	2.86	
Total Sensitivity | Certainty	**Low (1) | 0.75**		

Exposure

Exposure Factors	Mean	SD	Tally
Bottom Temperature	2.41	1.10	
Bottom Oxygen	2.49	1.05	
Bottom Salinity	3.15	0.73	
Bottom Aragonite Solubility	1.39	0.89	
Small zooplankton integrated biomass	1.3	0.39	
Medium zooplankton integrated biomass	1.32	0.37	
Large zooplankton integrated biomass	1.11	0.21	
Total Exposure	**Moderate (2.1)**	**0.97**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Low (2.1)**	**0.97**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Summer flounder. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 5 and Figure 4). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Jonathan A. Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	Low	Low	None
Exposure	Very High	Moderate	Decrease
Vulnerability	Moderate	Low	Decrease
Directional Effect	Neutral	Positive	Decrease

Stocks

Summer flounder is managed as a single stock or stock information was not considered for this iteration of the CVA.

Score Descriptions:

Sensitivity:

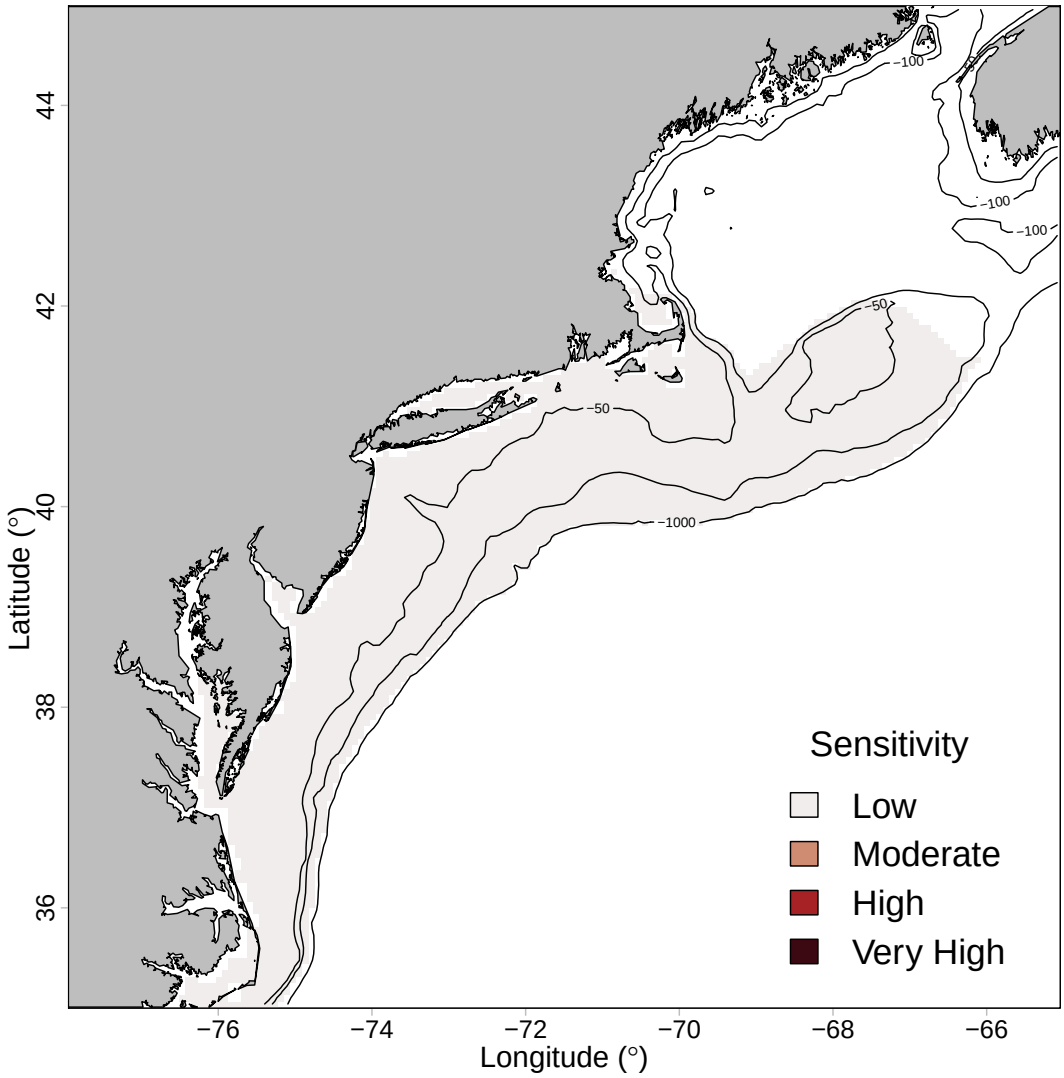


Figure 2: Summer flounder Sensitivity on the Northeast Shelf

Summary flounder received a **Low** sensitivity score. Certainty was 75% according to bootstrap analysis. The only attribute that received a score greater than 2.5 was Juvenile Habitat Requirements (2.7) (Figure 1). Juvenile summer flounder rely on estuary habitats, including marshes, oyster reefs, and seagrass beds for important nursery habitats. Many of these are considered highly vulnerable habitats (Guyant et al. 2022).

In the 2016 Climate Vulnerability Assessment, summer flounder also received a Low sensitivity score. In that assessment, no attributes scored above 2.5 (Jonathan A. Hare et al. 2016).

Exposure:

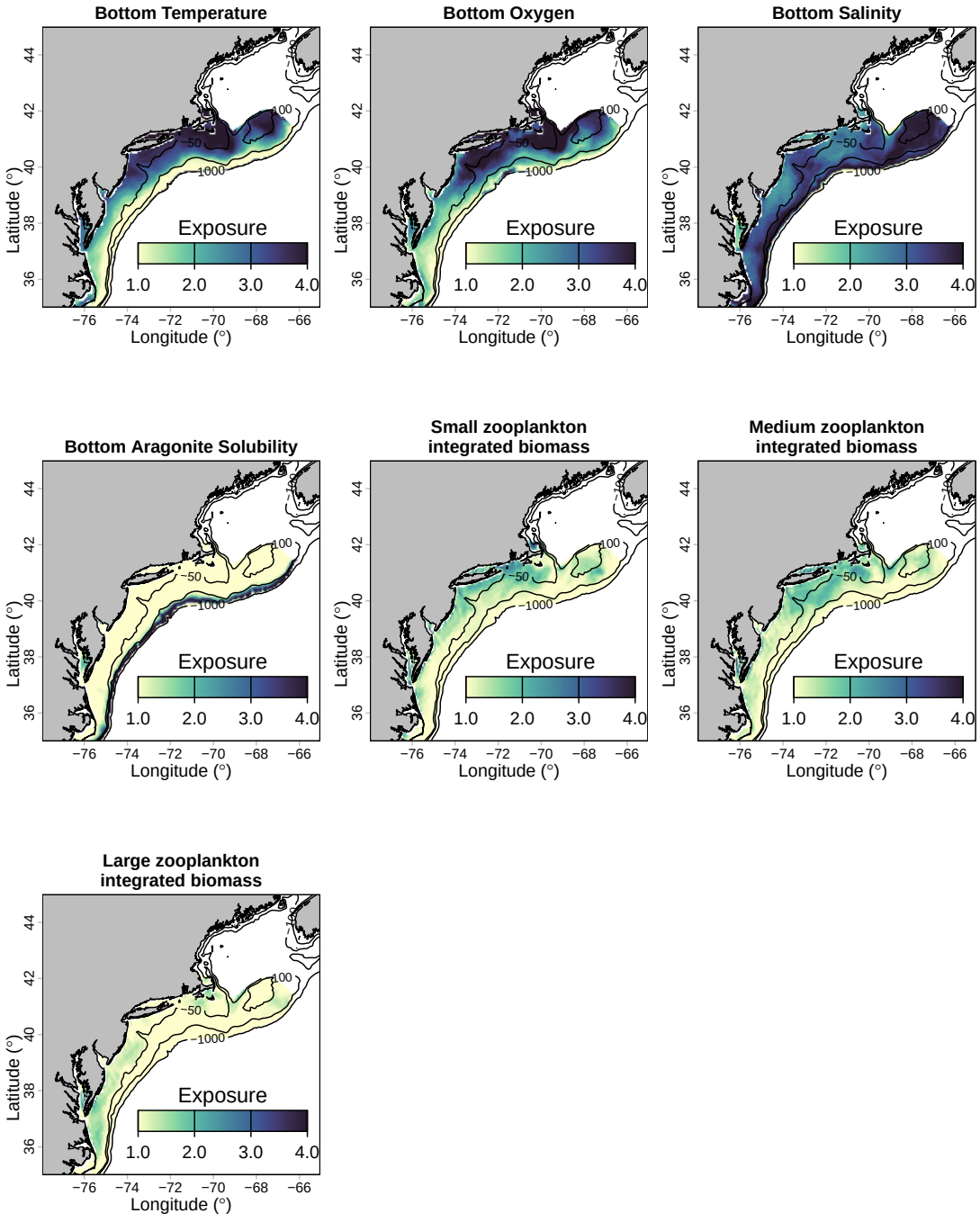


Figure 3: Maps of each Exposure Factor for Summer flounder. Exposure was calculated between 2014-2023 and 2025-2035

Mean exposure for Summer flounder was **Moderate** (2.1 +/- 0.97) (Figure 1). Exposure was Very High over Georges Bank and High throughout Southern New England and on the shelf break. Exposure was

Low on the shelf south of New Jersey (Figure 4). These patterns were driven by bottom temperature, oxygen, salinity, and aragonite solubility (Figure 3).

In the previous Assessment, summer flounder received a Very High exposure score. This was driven by high exposure to sea surface temperature, air temperature, and ocean acidification (Jonathan A. Hare et al. 2016).

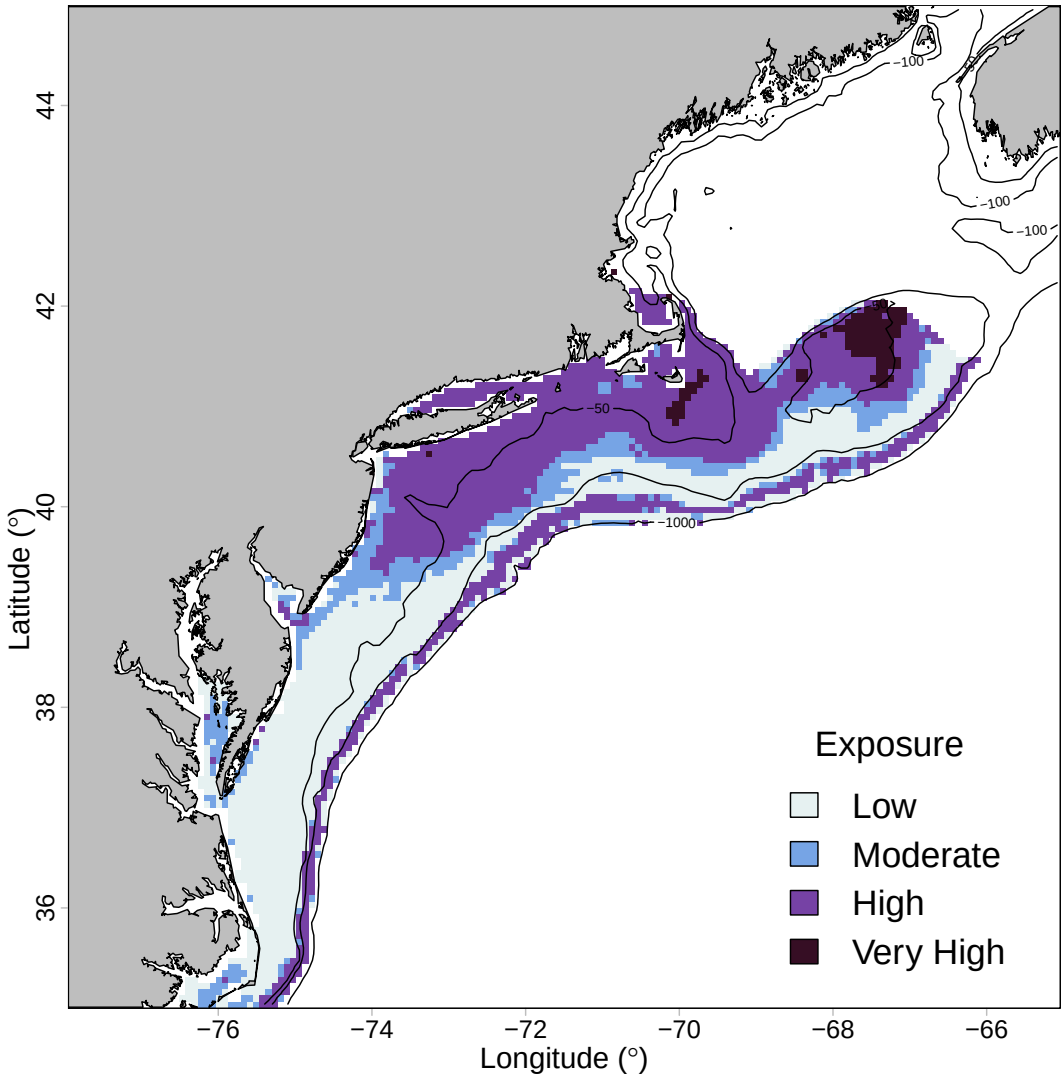


Figure 4: Total Exposure on the Northeast Shelf for Summer flounder. Total Exposure is calculated by applying the logic rule from Jonathan A. Hare et al. (2016) within grid cells across exposure factors (see Figure 3)

Vulnerability:

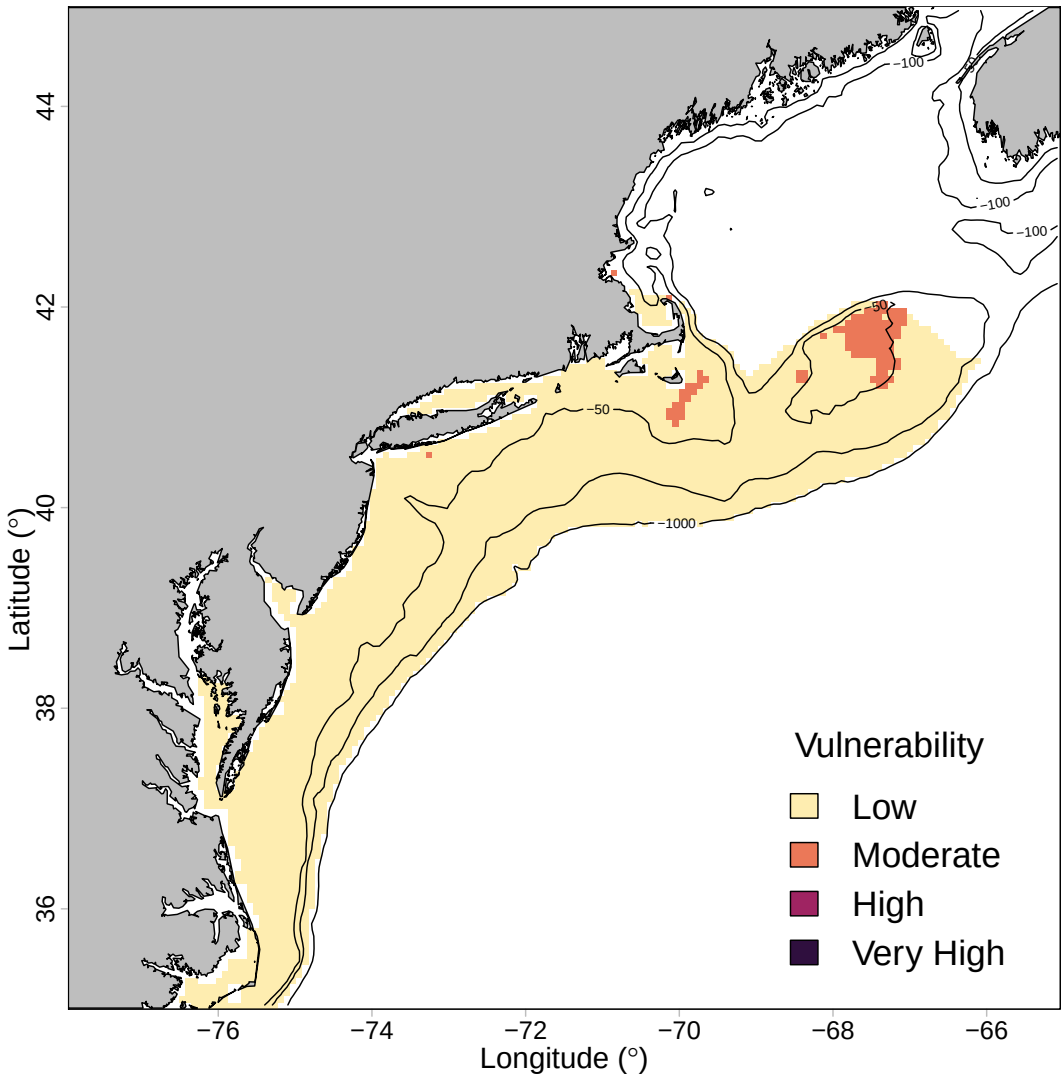


Figure 5: Vulnerability on the Northeast Shelf for Summer flounder. Vulnerability is the product of sensitivity and exposure (see Figure 4 and Figure 2)

Mean Vulnerability scores for Summer flounder were **Low** (2.1 +/- 0.97) (Figure 1). Vulnerability was Low throughout a majority of the summer flounder distribution, with the exception of portions of Georges Bank and areas south of Cape Cod (Figure 5).

In the previous Assessment, Summer Flounder received a Moderate Vulnerability score (Jonathan A. Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

100% of data quality scores were greater than or equal to 2, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were identified (Figure 1).

Model Confidence Score:

The summer flounder ensemble model received a mean model confidence score of 2 (+/- 0.6) indicating moderate confidence in the model. While the model predicted general distributions well, there were strong monthly patterns in residuals that drove scores down. The ensemble model AUC score was 0.94, indicating excellent model performance.

Additional Metrics

Distribution Shifts:

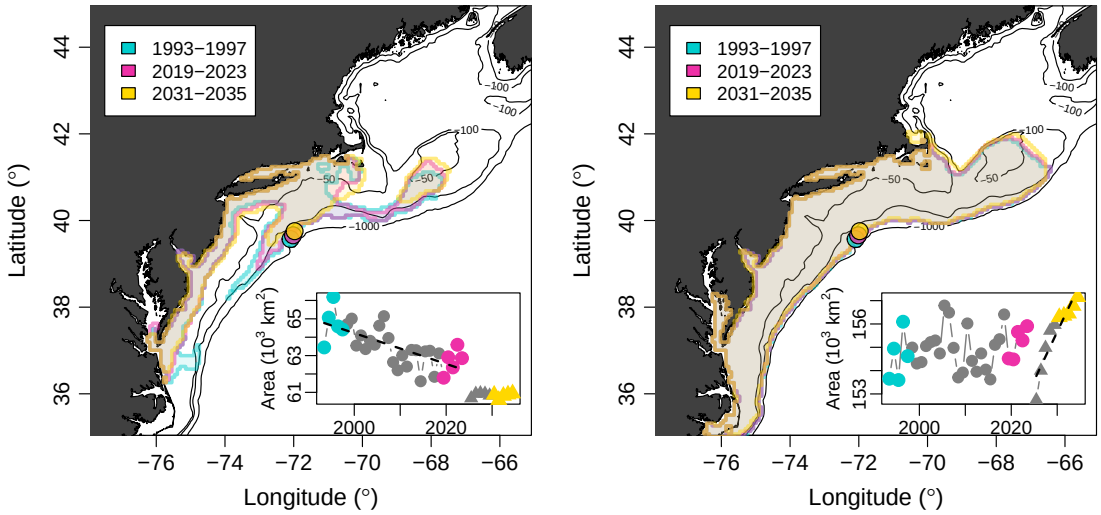


Figure 6: Distribution shifts on the Northeast Shelf for Summer flounder. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Summer flounder experienced significant core habitat loss between 1993 and 2023. These losses occurred at the southern edge of their range in both nearshore and shelfbreak habitats (Figure 6). The distribution centroid shifted 12 km northeast.

Under near future ocean conditions, summer flounder range is predicted to expand northward into Cape Cod Bay and further over Georges Bank (Figure 6). The distribution centroid is predicted to shift a further 11 km north-northeast.

In the previous assessment, summer flounder received a High Distributional Vulnerability rank due to high larval dispersal, seasonal migrations, and variety of preferred habitats (Jonathan A. Hare et al. 2016).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	65,140	154,589	5.11	8.10
2019-2023	11.5 km NE	62,882 (-2,258)	155,202 (+613)	4.99 (-0.12)	8.10 (0)
2031-2035	11.3 km NNE	60,800 (-2,082)	156,586 (+1,384)	4.93 (-0.06)	8.10 (0)

Directional Effect:

Experts predict that summer flounder will be **positively** impacted by climate change. Certainty was 65%. Spatio-temporal models have suggested that summer flounder abundance will increase with increasing temperatures (Deen, Jauss, and Sullivan 2025). Increasing winter water temperatures may help summer flounder expand their distributions northward (Bell et al. 2014). In the previous Assessment, experts predicted that the impacts of climate change on summer flounder would be neutral; however, certainty was low (Jonathan A. Hare et al. 2016).

Climate Effects on Abundance and Distribution

Summer flounder distributions are correlated with depth, bottom temperature, sea surface temperature (SST), and bottom salinity (Deen, Jauss, and Sullivan 2025). Their distribution in the Northwest Atlantic may shifting north due to increasing winter temperatures, but could also be affected by reductions in fishing pressure or range expansion (Bell et al. 2014). Large-scale climate drivers, such as the Gulf Stream Index, may affect summer flounder recruitment and distribution (O'Leary et al. 2019). Spatio-temporal modeling results from (Deen, Jauss, and Sullivan 2025) report increased abundance of summer flounder in the fall by 10-15% with a 1°C increase in SST, 12% with a 1°C increase in bottom temperature and 3-4% increase with each meter in depth. In the spring, abundance increased by 30% with a 1°C increase in SST. Summer flounder embryos exhibit reduced survival when exposed to elevated CO^2. Larvae exposed to elevated CO^2 had tissue damage and initiated metamorphosis earlier and at smaller sizes (Chambers et al. 2014).

Life History Synopsis

Summer flounder are found both inshore and offshore in waters from Nova Scotia to Florida (Packer et al. 1999). Spawning season lasts from September through February/March, with lower latitudes starting sooner. Spawning adults migrate to overwintering habitats on the continental shelf edge. In the late spring, they move back inshore into shallow waters and estuaries (Packer et al. 1999). The timing of peak spawning in October/November coincides with the breakdown of thermal stratification on the continental

shelf and the maximum production of autumn plankton which is characteristic of temperate ocean waters of the northern hemisphere (Packer et al. 1999) (Morse 1981). Egg development takes anywhere from 48-142 hours, depending on water temperatures. At 14°C, larvae metamorphosize into juveniles within approximately 65 days of hatching. Time to metamorphosis also was dependent on temperatures, with metamorphosis completing as fast as 46.5 days at 14.5°C (Smith 1973). Larvae do not drift far from spawning regions; however, tidal stream transport and vertical migration behavior may be important to determining larval distributions (Packer et al. 1999) (J. A. Hare, Walsh, and Wuenschel 2006). Larvae prefer waters with warmer temperatures and intermediate salinities. Temperature affects the rate of larvae yolk sac absorption (which impacts when larvae have to depend on exogenous feeding), development and growth (Watanabe et al. 1998). Studies have also shown that larval size increases with greater light intensity (presumably affecting activity and energy metabolism) and higher salinities (Watanabe et al. 1998) (Burke et al. 1998). Larvae are preyed upon by a variety of predators, including blue crabs and sea robins. The type and abundance of predators (ex: *Crangon septemspinosa* or pelagic mummichugs) could determine whether a metamorphic summer flounder stays in the substrate or the water column (*C. septemspinosa* would force them to the water column, and vice versa for mummichugs) (Keefe and Able 1994). Larvae preferentially settle on sandy substrate; bottom type, presence of predators, and water temperatures may all impact settlement (Keefe and Able 1994). Young-of-the-year can occur in a variety of habitats, including shallow, mud-bottomed marsh creeks, shallow sand substrates (including seagrass beds), deep sand substrate, and deep fine-sand substrates (Szedlmayer and Able 1993). Juveniles are opportunistic feeders and differences in diet are often related to the availability of prey. There also appears to be ontogenetic changes in diet. Smaller individuals eat polychaetes and crustaceans. Metamorphic stage individuals reflect calanoid copepod diets (pelagic), while late metamorphosis stage individual and juveniles' diets reflected shrimp, polychaetes, harpactoid copepods, amphipods, small fishes, small gastropod mollusks, and plant material (benthic) (Grover 1998). Adults are demersal, with highest abundance in waters deeper than 30m and at temperatures 2-27°C (Packer et al. 1999). Adults are also opportunistic feeders. Fish and crustaceans make up the majority of their diet, but dominant prey items shift following prey availability (Cernadas-Martín et al. 2021). Adult summer flounder are preyed upon by sharks, rays, and monkfish (Packer et al. 1999). Summer flounder reach maturity at approximately 13-17 months. Males can reach 12 years of age and females 14 years, although individuals living upwards of 20 years have been reported (Northeast Fisheries Science Center (NEFSC) 2019). The summer flounder stock is cooperatively managed by the Mid-Atlantic Fishery Management Council and Atlantic States Marine Fisheries Commission under the Summer Founder, Scup, and Black Sea Bass Fishery Management Plan. The stock is not overfished and overfishing is not occurring (Northeast Fisheries Science Center (NEFSC) 2025).

Supplemental Scores

Exposure Timeseries

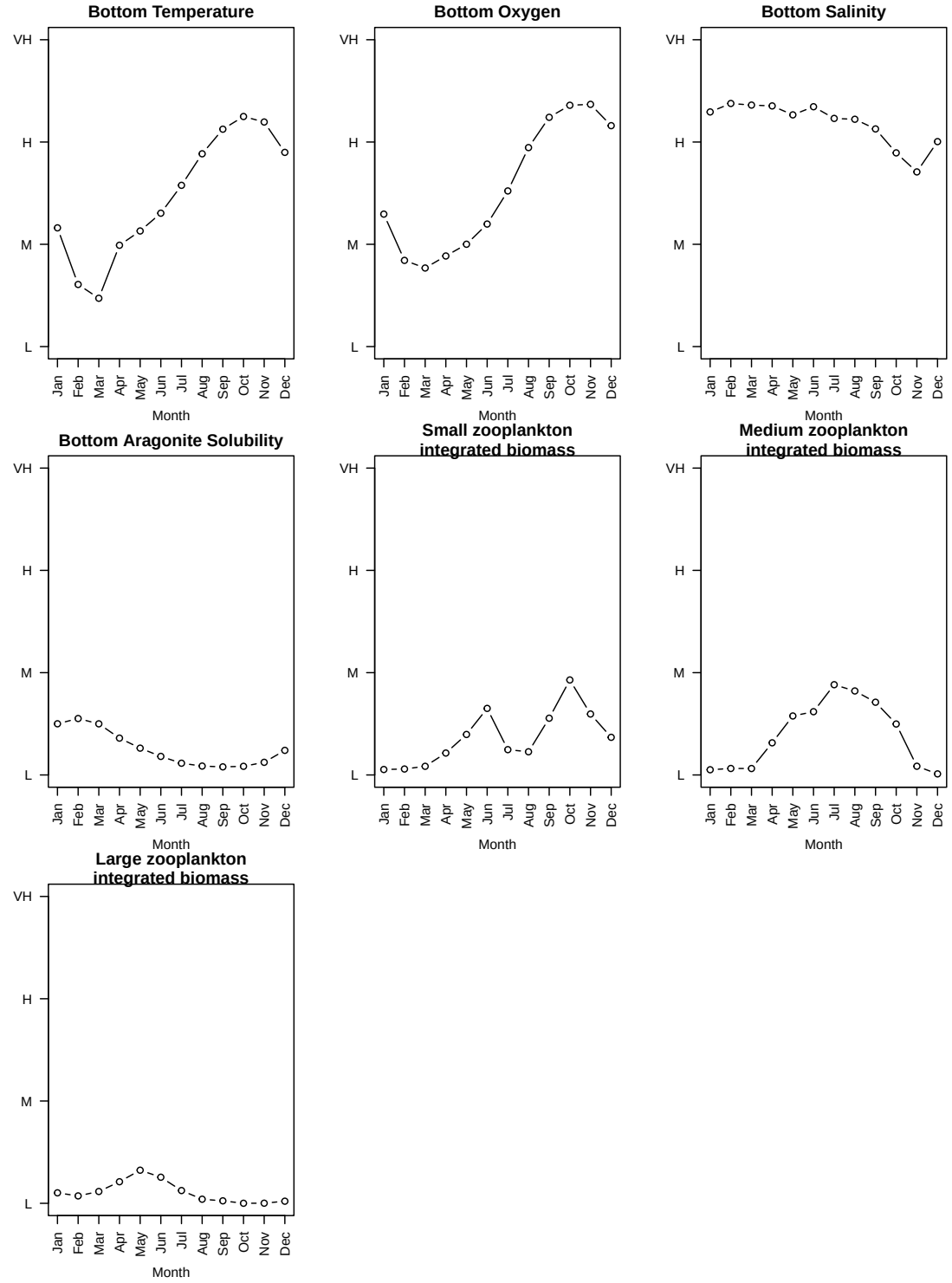


Figure 7: Monthly Timeseries of each Exposure Factor for Summer flounder. Exposure was calculated between 2014-2023 and 2025-2035

Monthly exposure ranged from Low to High, with the highest scores occurring between September through December (Figure 8). This was driven by peaks in bottom temperature and bottom oxygen exposure during these months (Figure 7).

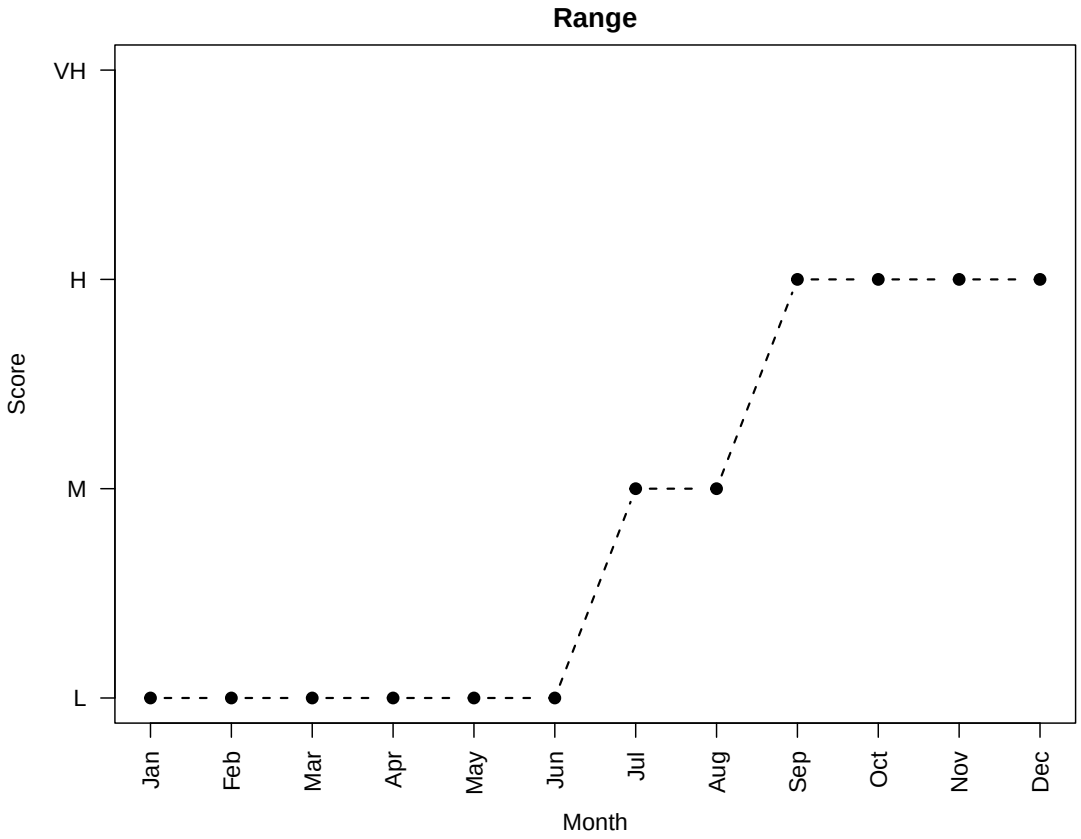


Figure 8: Timeseries of Total Exposure for Summer flounder

Exposure due to Significant Factors

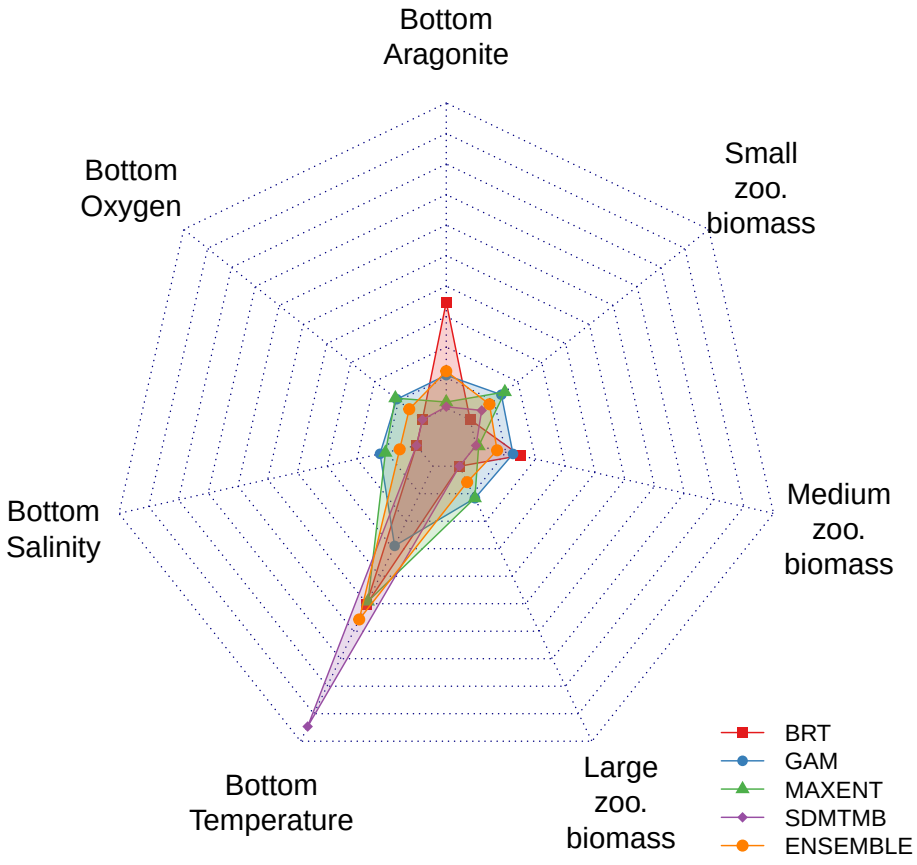


Figure 9: Dynamic Factors driving Summer flounder distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Bottom temperature and aragonite solubility were the major drivers of summer flounder distributions (Figure 9). Adults are demersal, opportunistic feeders (Packer et al. 1999) (Cernadas-Martín et al. 2021). When only these factors were used to calculate exposure, scores decreased to Low (1.01 +/- 0.11) (Figure 11, Figure 10).

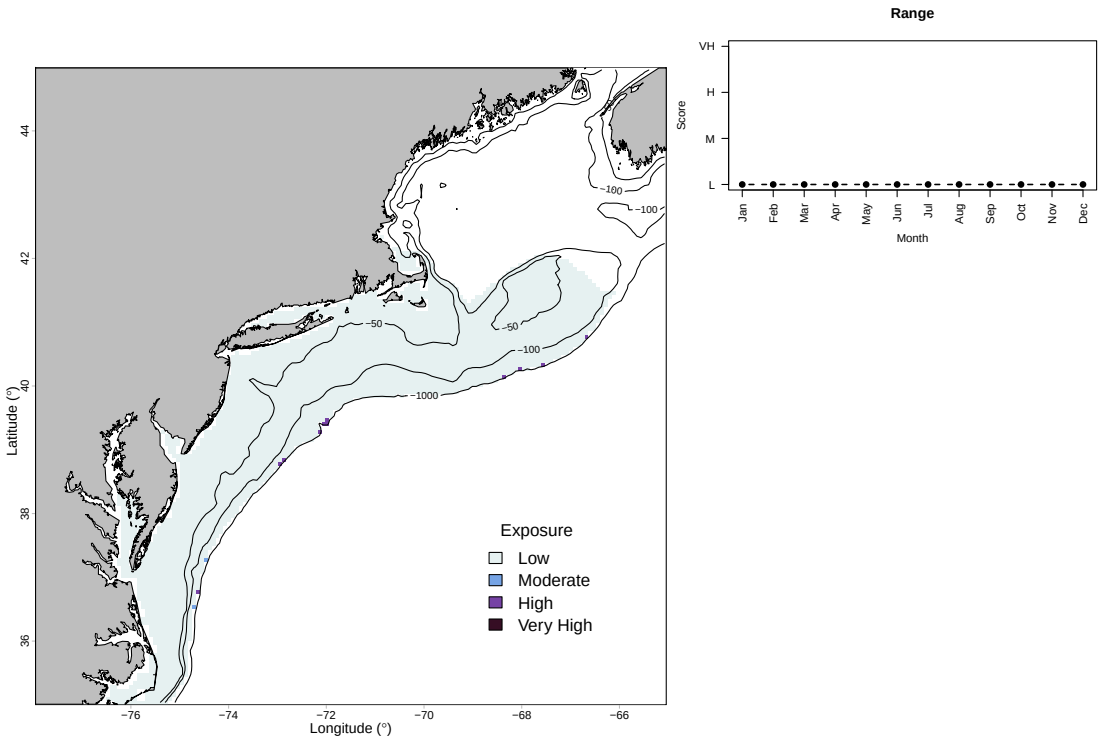


Figure 10: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Summer flounder distributions

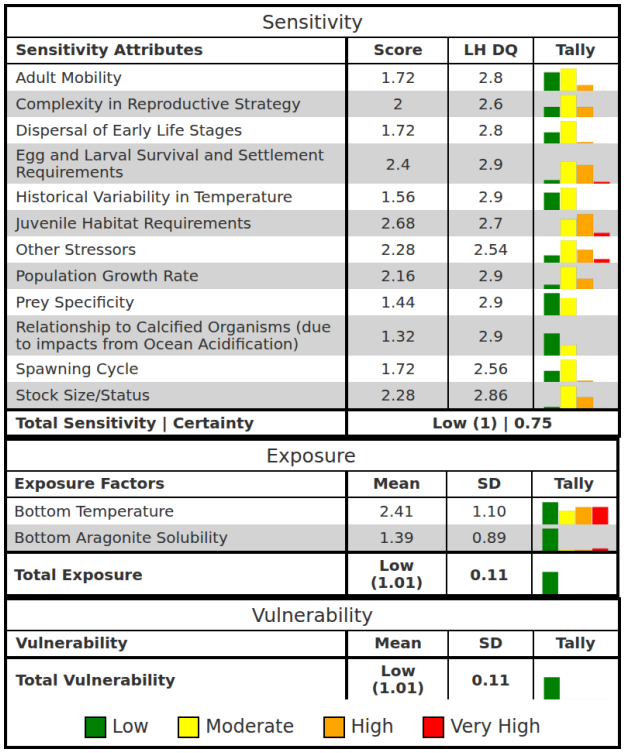


Figure 11: As in Figure 1 for Summer flounder, but with only significant Exposure factors and the resulting Vulnerability Scores.

References

Bell, R. J., J. A. Hare, J. P. Manderson, and D. E. Richardson. 2014. "Externally Driven Changes in the Abundance of Summer and Winter Flounder." *ICES Journal of Marine Science* 71 (9): 2416–28.

Burke, J. S., M. Ueno, Y. Tanaka, H. Walsh, T. Maeda, I. Kinoshita, T. Seikai, D. E. Hoss, and M. Tanaka. 1998. "The Influence of Environmental Factors on Early Life History Patterns of Flounders." *Journal of Sea Research* 40 (1-2): 19–32.

Cernadas-Martín, S., K. J. Rountos, J. A. Nye, M. G. Frisk, and E. K. Pikitch. 2021. "Composition and Intraspecific Variability in Summer Flounder (Paralichthys dentatus) Diets in a Eutrophic Estuary." *Frontiers in Marine Science* 8: 632751.

Chambers, R. C., A. C. Candelmo, E. A. Habeck, M. E. Poach, D. Wieczorek, K. R. Cooper, C. E. Greenfield, and B. A. Phelan. 2014. "Effects of Elevated CO_2 in the Early Life Stages of Summer Flounder, Paralichthys dentatus, and Potential Consequences of Ocean Acidification." *Biogeosciences* 11 (6): 1613–26.

Deen, S., V. Jauss, and P. J. Sullivan. 2025. "Estimating Effects of Ocean Environmental Conditions on Summer Flounder (Paralichthys dentatus) Distribution." *Environmental and Ecological Statistics* 32 (1): 283–310.

Grover, J. J. 1998. "Feeding Habits of Pelagic Summer Flounder, (Paralichthys dentatus), Larvae in Oceanic and Estuarine Habitats." *Fishery Bulletin* 96: 248–57.

Guyant, M, M Bachman, J Coakley, G Fay, M Johnson, V Kentner, W Morrison, and T Rankin. 2022. "Northeast Regional Habitat Assessment-Habitat Climate Vulnerability Assessment Crosswalk, Including Species Narratives, Habitat-Species Matrix, and Final Report." Northeast Regional Habitat Assessment. https://nrha.shinyapps.io/dataexplorer/#!/crosswalk.

Hare, J. A., H. J. Walsh, and M. J. Wuenschel. 2006. "Sinking Rates of Late-Stage Fish Larvae: Implications for Larval Ingress into Estuarine Nursery Habitats." *Journal of Experimental Marine Biology and Ecology* 330 (2): 493–504.

Hare, Jonathan A, Wendy E Morrison, Mark W Nelson, Megan M Stachura, Eric J Teeters, Roger B Griffis, Michael A Alexander, et al. 2016. "A Vulnerability Assessment of Fish and Invertebrates to Climate Change on the Northeast US Continental Shelf." *PloS One* 11 (2): e0146756.

Keefe, M., and K. W. Able. 1994. "Contributions of Biotic Factors to Settlement in Summer Flounder, Paralichthys dentatus." *Copeia* 1994 (2): 458–65.

Morse, W. W. 1981. "Reproduction of the Summer Flounder, Paralichthys dentatus (L.)." *Journal of Fish Biology* 19: 189–203.

Northeast Fisheries Science Center (NEFSC). 2019. "66th Northeast Regional Stock Assessment Workshop (66th SAW) Assessment Report." 19-08. National Marine Fisheries Service. https://doi.org/10.25923/nhqe-jd35.

———. 2025. "Summer Flounder 2025 Management Track Assessment Report." Atlantic States Marine Fisheries Commission. https://asmfc.org/wp-content/uploads/2025/08/SF_Management_Track_Assessment_2025.pdf.

O'Leary, C. A., T. J. Miller, J. T. Thorson, and J. A. Nye. 2019. "Understanding Historical Summer Flounder (Paralichthys dentatus) Abundance Patterns Through the Incorporation of Oceanography-Dependent Vital Rates in Bayesian Hierarchical Models." *Canadian Journal of Fisheries and Aquatic Sciences* 76 (8): 1275–94.

Packer, D. B., S. J. Griesbach, P. B. Berrien, C. A. Zetlin, D. L. Johnson, and W. W. Morse. 1999. "Essential Fish Habitat Source Document. Summer Flounder, Paralichthys dentatus, Life History and Habitat Characteristics." NOAA Technical Memorandum NMFS-NE 151. National Marine Fisheries Service.

Smith, W. G. 1973. "The Distribution of Summer Flounder, Paralichthys dentatus, Eggs and Larvae on the Continental Shelf Between Cape Cod and Cape Lookout, 1965-66." *Fishery Bulletin* 71: 527–48.

Szedlmayer, S. T., and K. W. Able. 1993. "Ultrasonic Telemetry of Age-0 Summer Flounder, Paralichthys dentatus, Movements in a Southern New Jersey Estuary." *Copeia* 1993 (3): 728–36.

Watanabe, W. O., M. W. Feeley, S. C. Ellis, and E. P. Ellis. 1998. "Light Intensity and Salinity Effects on Eggs and Yolk Sac Larvae of the Summer Flounder." *North American Journal of Aquaculture* 60 (1): 9–19.